Supporting Information

Dawn of the dinosaurs demonstrates early diversification bursts and potential role for neutral dynamics during major evolutionary radiations

Roland B. Sookias, Luciano A. Pradelli, Eduardo Ascarrunz, Daniele Silvestro, Martín D. Ezcurra

Correspondence: sookias.r.b@gmail.com, m.d.ezcurra@bham.ac.uk

Journal: Proceedings of the Royal Society B

DOI: 10.1098/rspb.2026.0485

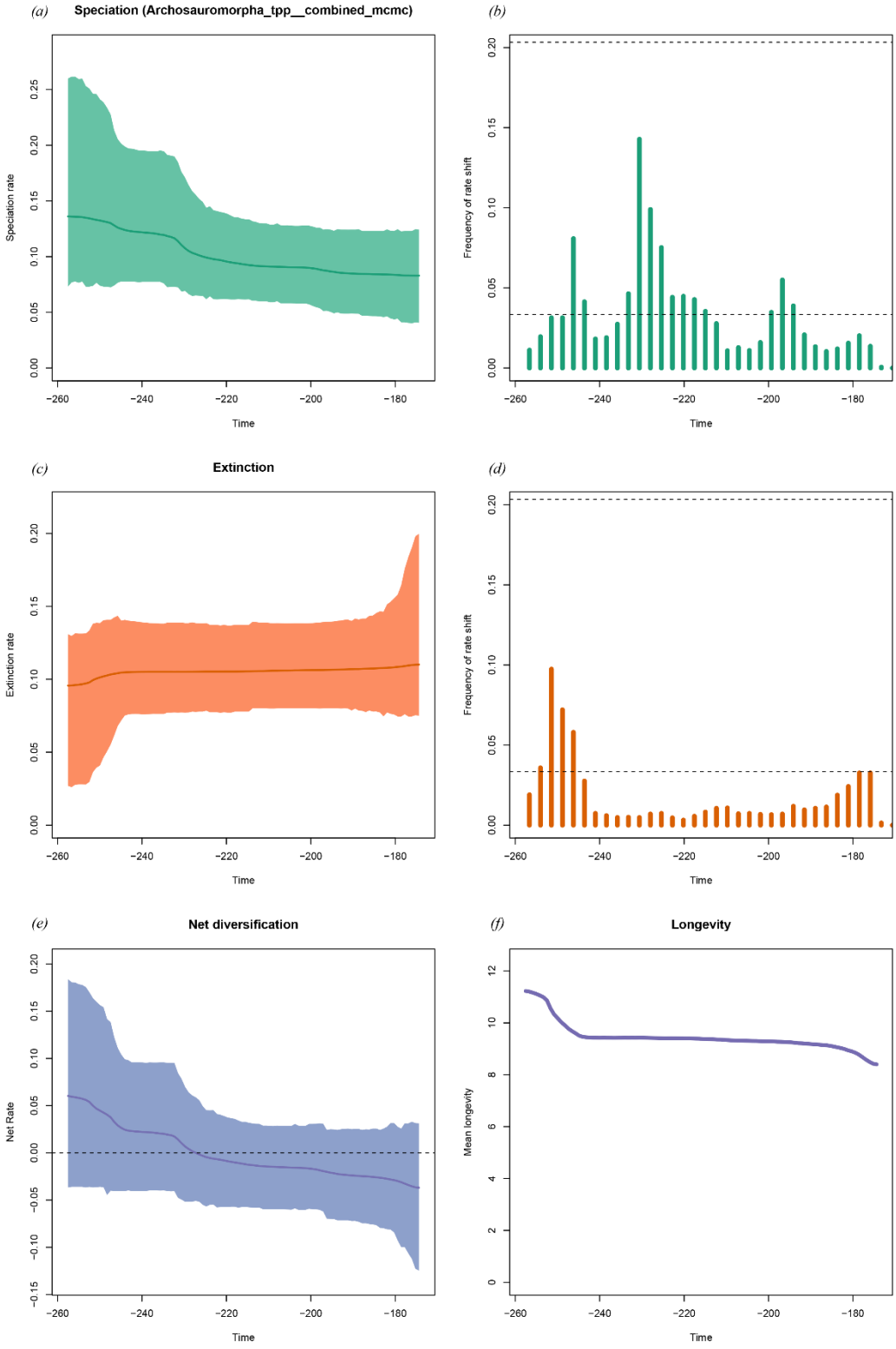


Figure S-1. Diversification dynamics curves, shifts, and taxon longevities estimated with the TPP model. Left plots show mean rates and 95% confidence interval, upper two right plots show frequencies of significant rate shifts, and lower right plot show mean longevity of taxa through time. Speciation (green) and extinction (red) rates are per lineage per Myr, and the diversification rate (purple) is the difference between speciation and extinction rates.

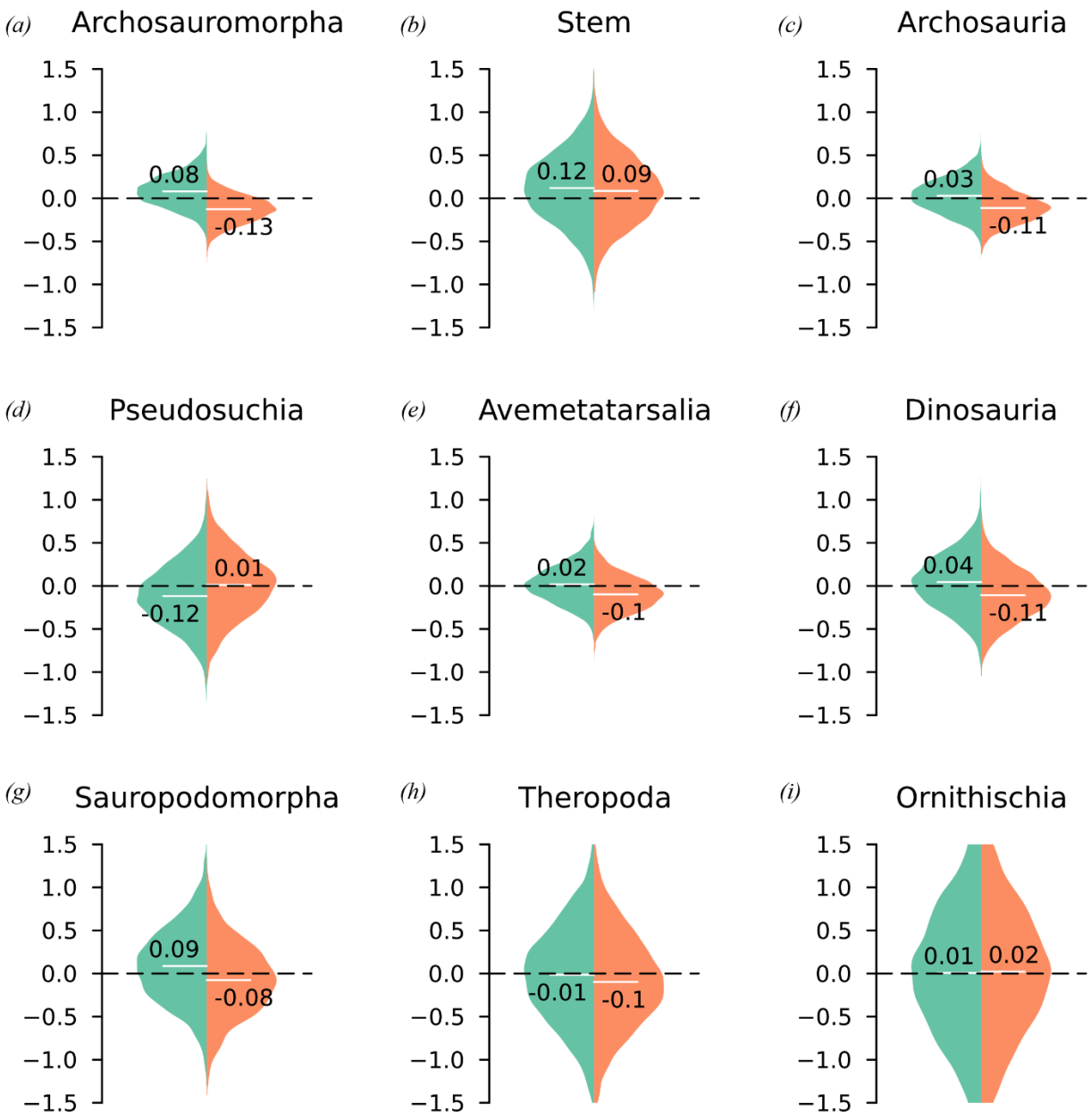


Figure S-2. Posterior probability densities of covariances between log10-femoral length and speciation (green) or extinction (red) rates for Archosauromorpha and its subclades.

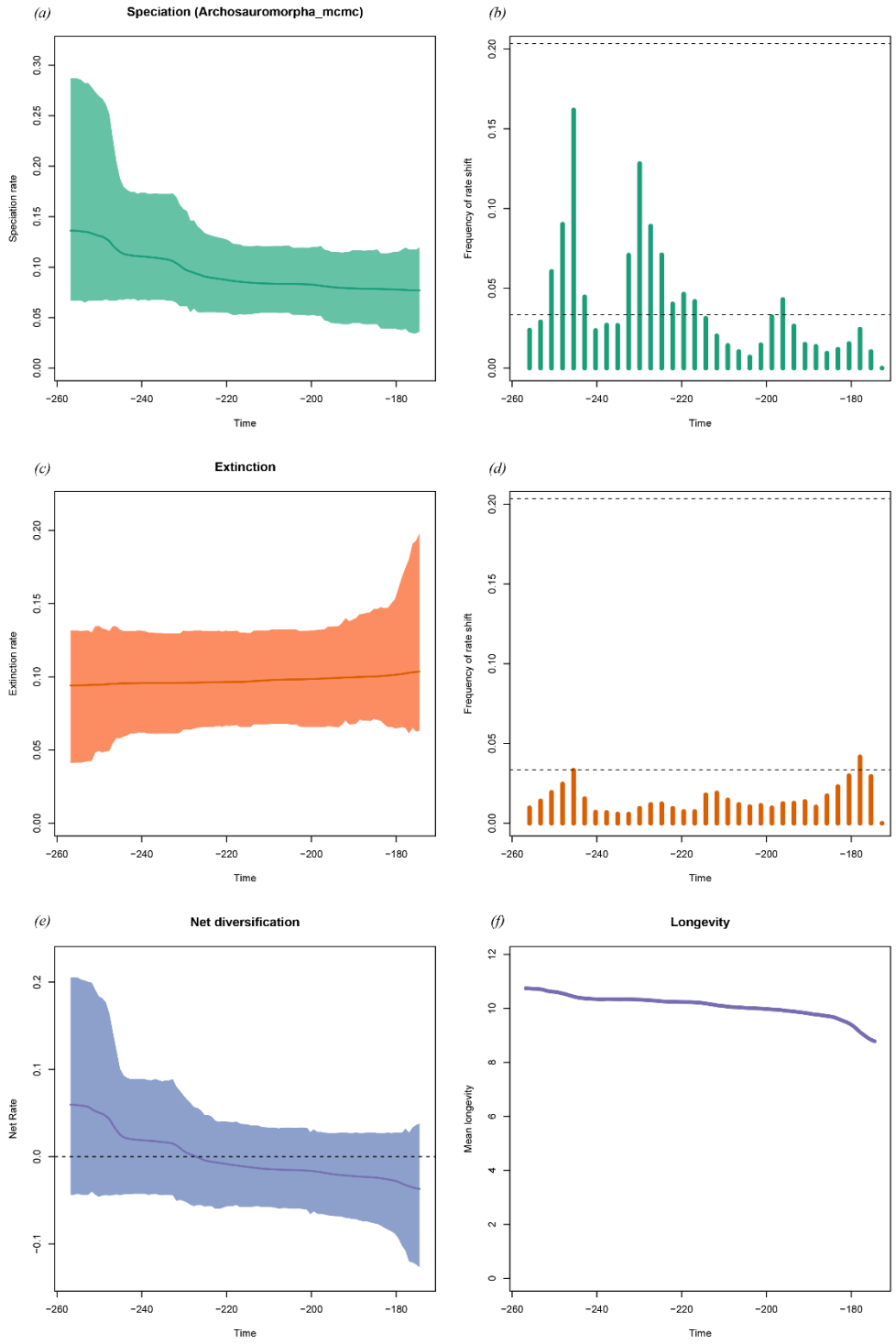


Figure S-3. Full diversification dynamics curves, shifts, and taxon longevities estimated as outputted by PyRate with the HPP model for Archosauromorpha. Left plots show mean rates and 95% confidence interval, upper two right plots show frequencies of significant rate shifts, and lower right plot shows mean longevity of taxa through time. Speciation (green) and extinction (red) rates are per lineage per Myr, and the diversification rate (purple) is the difference between speciation and extinction rates.

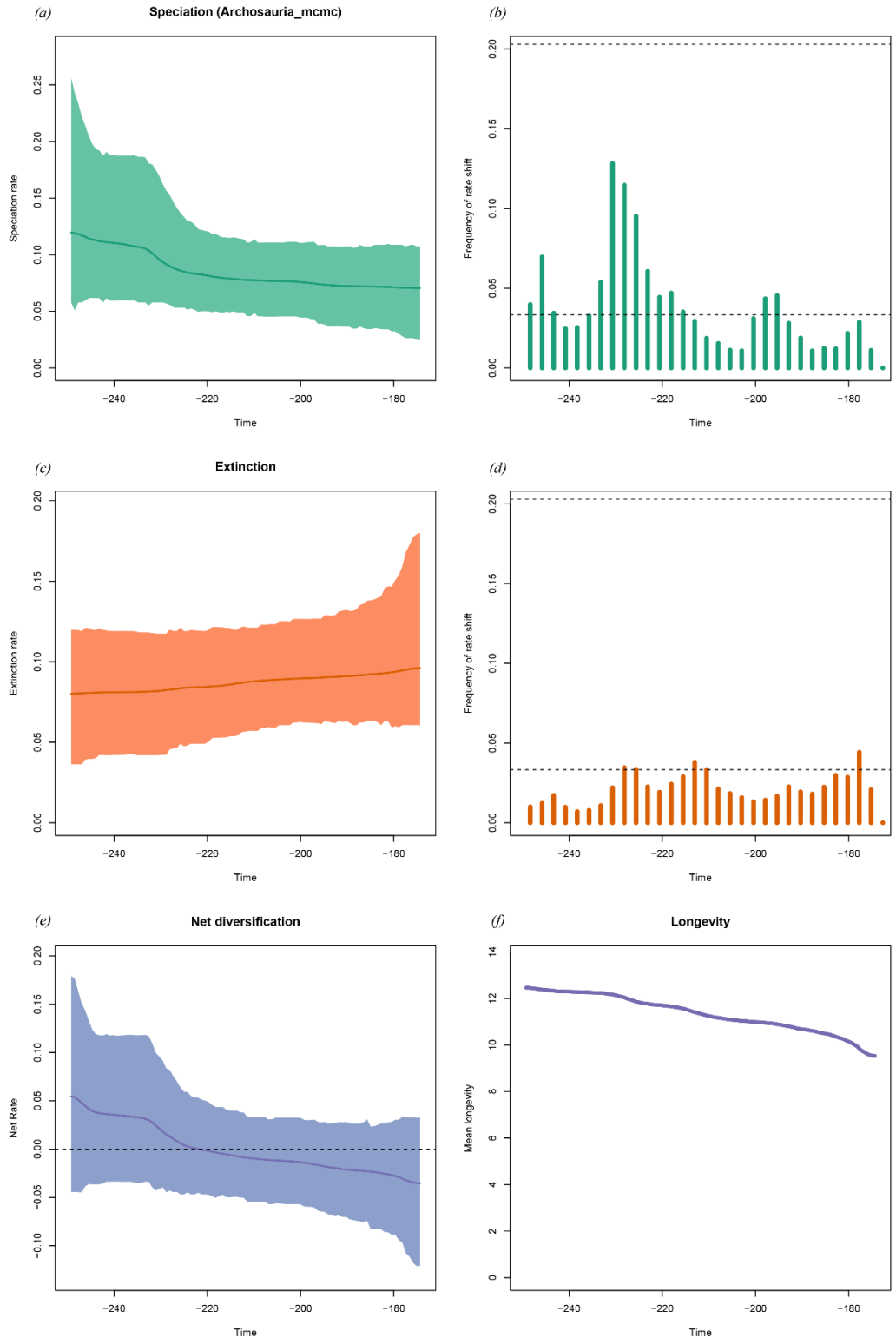


Figure S-4. Full diversification dynamics curves, shifts, and taxon longevities estimated as outputted by PyRate with the HPP model for Archosauria. Left plots show mean rates and 95% confidence interval, upper two right plots show frequencies of significant rate shifts, and lower right plot shows mean longevity of taxa through time. Speciation (green) and extinction (red) rates are per lineage per Myr, and the diversification rate (purple) is the difference between speciation and extinction rates.

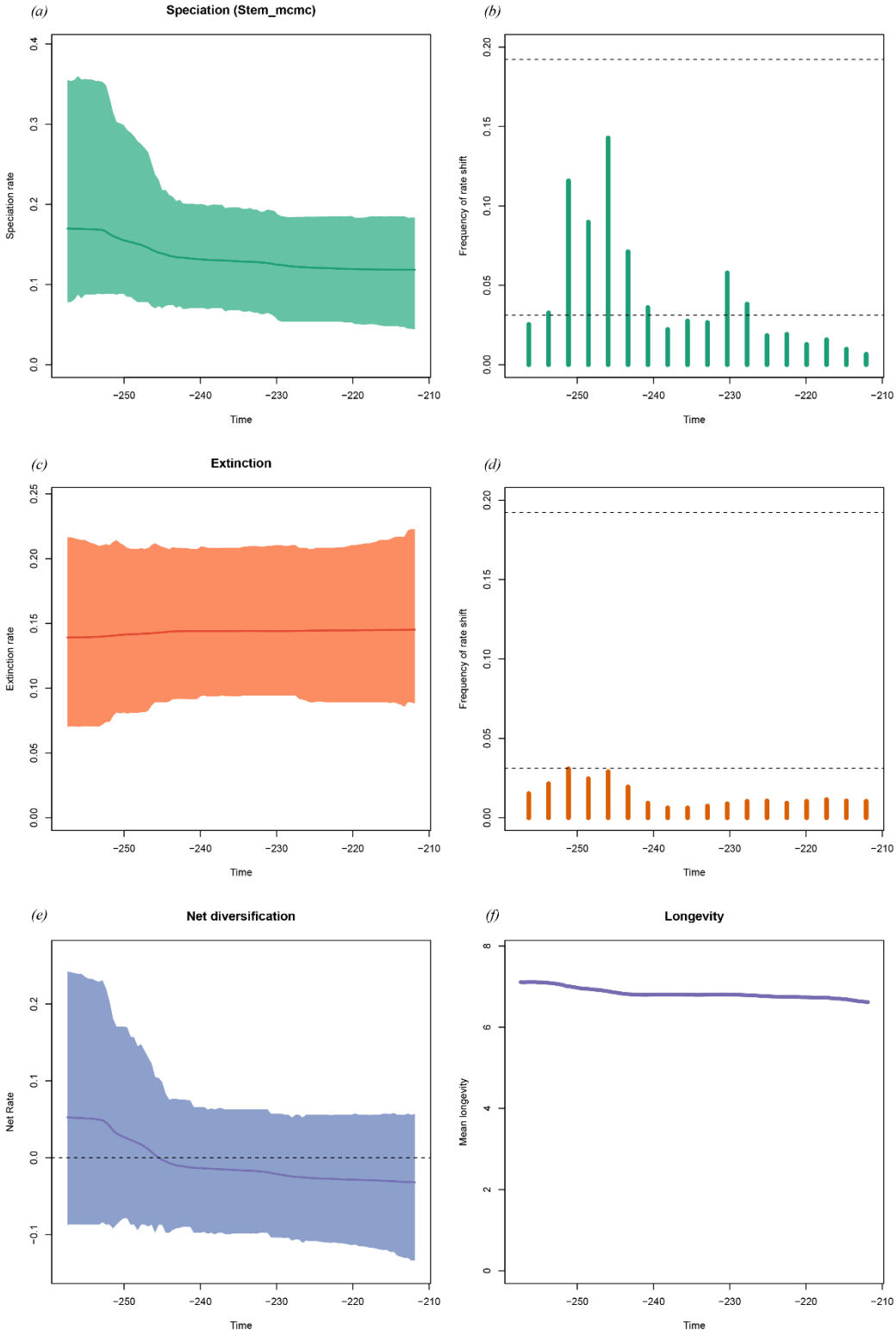


Figure S-5. Full diversification dynamics curves, shifts, and taxon longevities estimated as outputted by PyRate with the HPP model for stem Archosauria. Left plots show mean rates and 95% confidence interval, upper two right plots show frequencies of significant rate shifts, and lower right plot shows mean longevity of taxa through time. Speciation (green) and extinction (red) rates are per lineage per Myr, and the diversification rate (purple) is the difference between speciation and extinction rates.

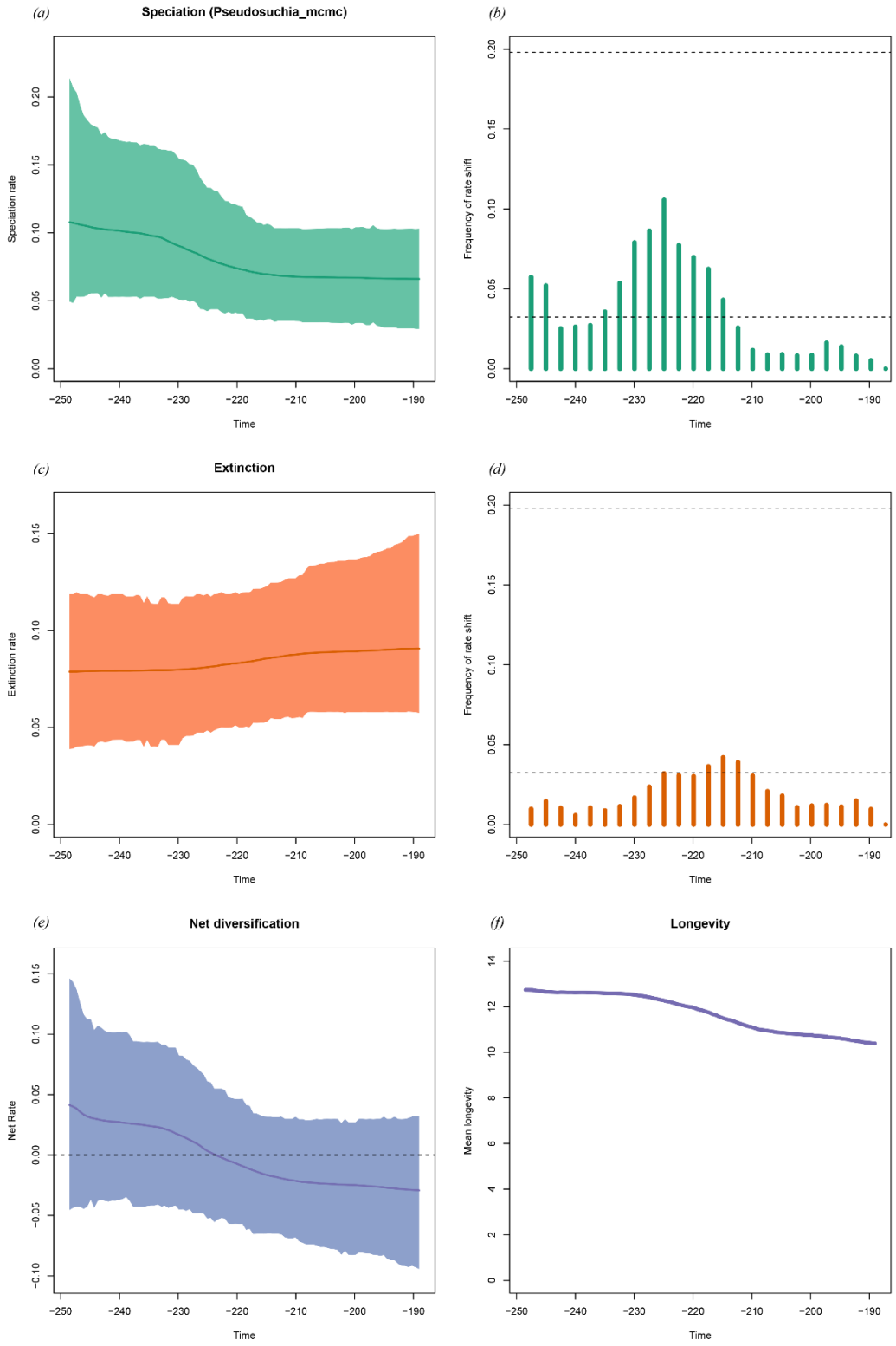


Figure S-6. Full diversification dynamics curves, shifts, and taxon longevities estimated as outputted by PyRate with the HPP model for Pseudosuchia. Left plots show mean rates and 95% confidence interval, upper two right plots show frequencies of significant rate shifts, and lower right plot shows mean longevity of taxa through time. Speciation (green) and extinction (red) rates are per lineage per Myr, and the diversification rate (purple) is the difference between speciation and extinction rates.

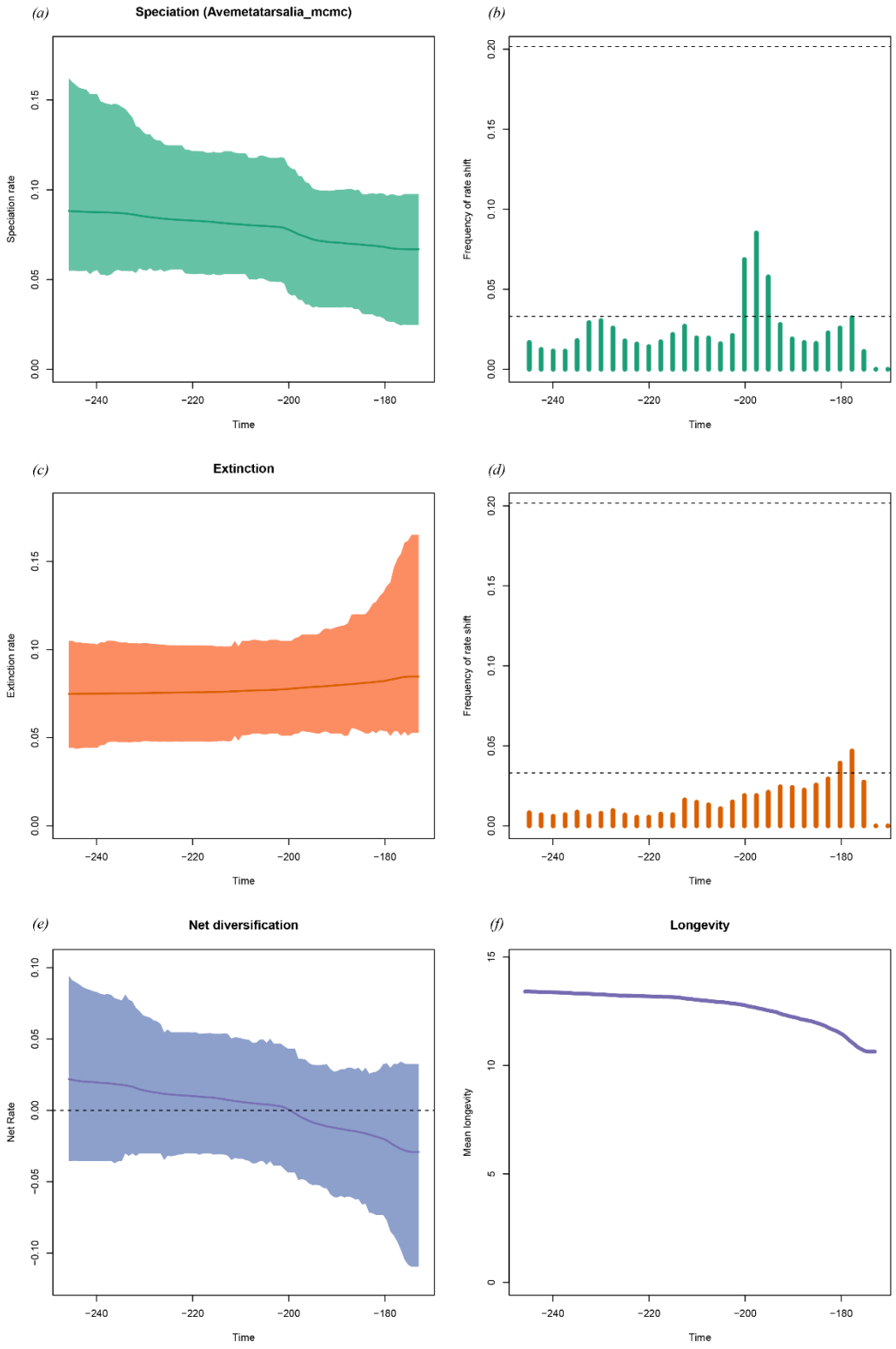


Figure S-7. Full diversification dynamics curves, shifts, and taxon longevities estimated as outputted by PyRate with the HPP model for Avemetatarsalia. Left plots show mean rates and 95% confidence interval, upper two right plots show frequencies of significant rate shifts, and lower right plot shows mean longevity of taxa through time. Speciation (green) and extinction (red) rates are per lineage per Myr, and the diversification rate (purple) is the difference between speciation and extinction rates.

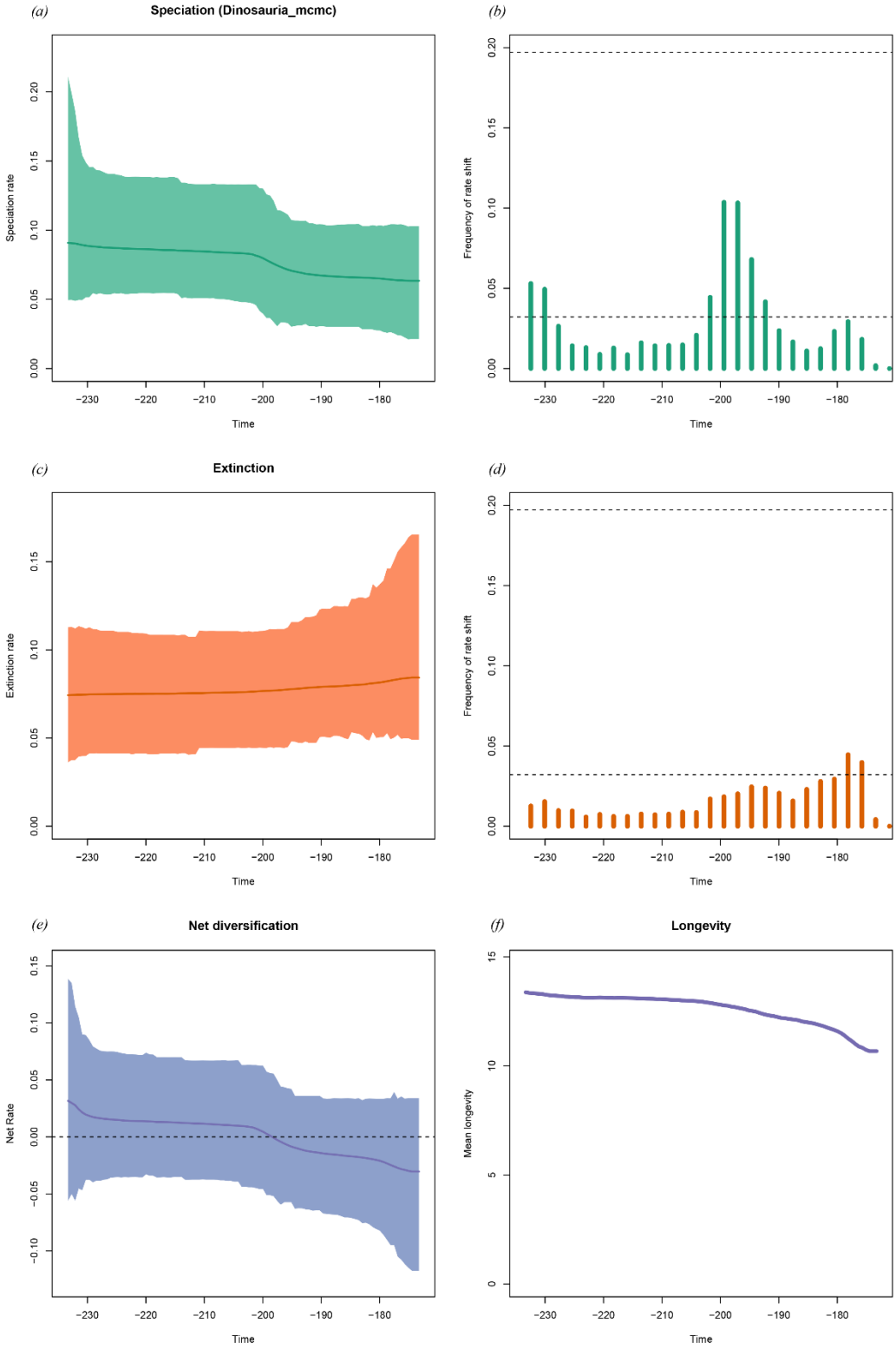


Figure S-8. Full diversification dynamics curves, shifts, and taxon longevities estimated as outputted by PyRate with the HPP model for Dinosauria. Left plots show mean rates and 95% confidence interval, upper two right plots show frequencies of significant rate shifts, and lower right plot shows mean longevity of taxa through time. Speciation (green) and extinction (red) rates are per lineage per Myr, and the diversification rate (purple) is the difference between speciation and extinction rates.

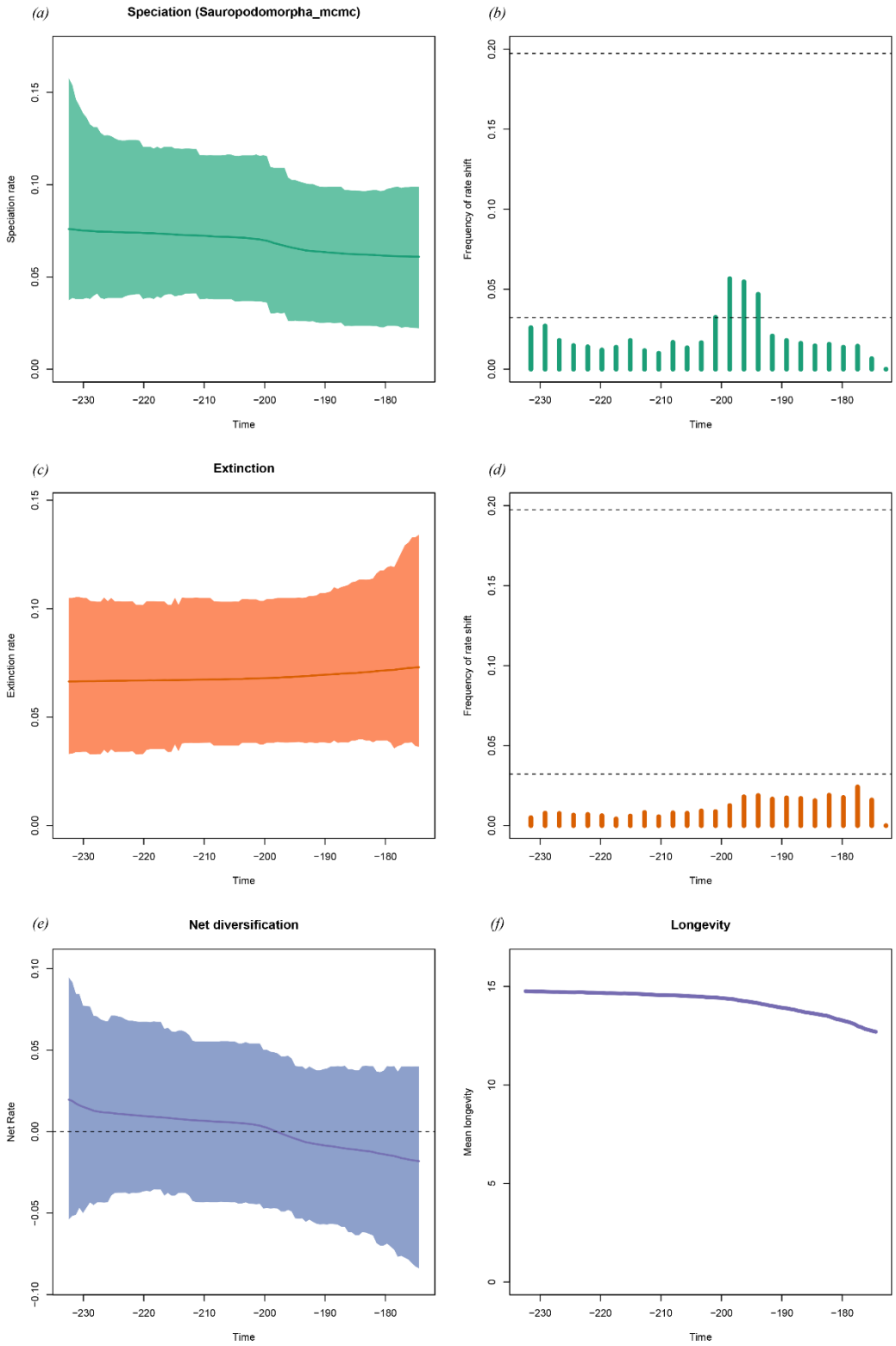


Figure S-9. Full diversification dynamics curves, shifts, and taxon longevities estimated as outputted by PyRate with the HPP model for Sauropodomorpha. Left plots show mean rates and 95% confidence interval, upper two right plots show frequencies of significant rate shifts, and lower right plot shows mean longevity of taxa through time. Speciation (green) and extinction (red) rates are per lineage per Myr, and the diversification rate (purple) is the difference between speciation and extinction rates.

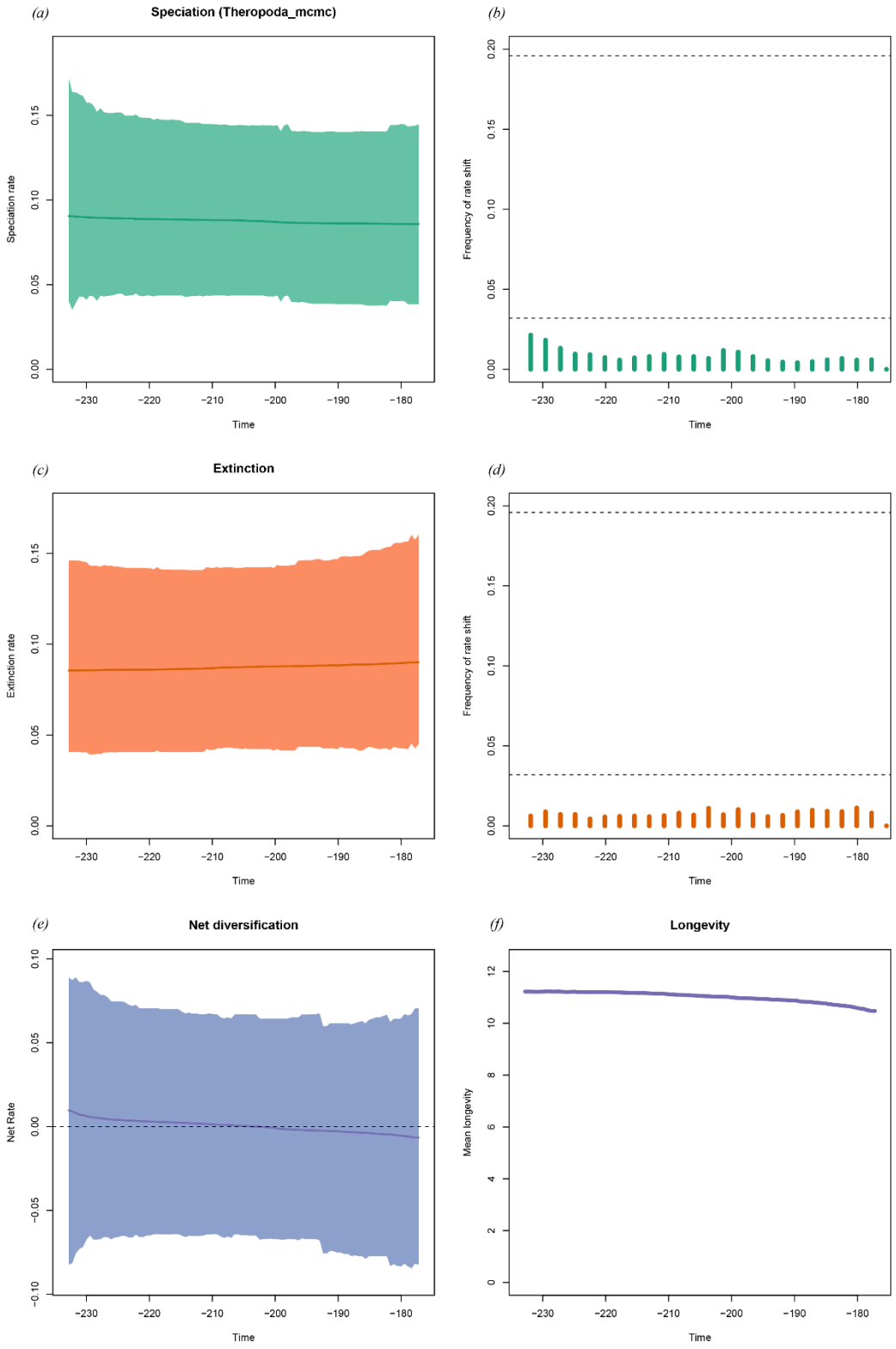


Figure S-10. Full diversification dynamics curves, shifts, and taxon longevities estimated as outputted by PyRate with the HPP model for Theropoda. Left plots show mean rates and 95% confidence interval, upper two right plots show frequencies of significant rate shifts, and lower right plot shows mean longevity of taxa through time. Speciation (green) and extinction (red) rates are per lineage per Myr, and the diversification rate (purple) is the difference between speciation and extinction rates.

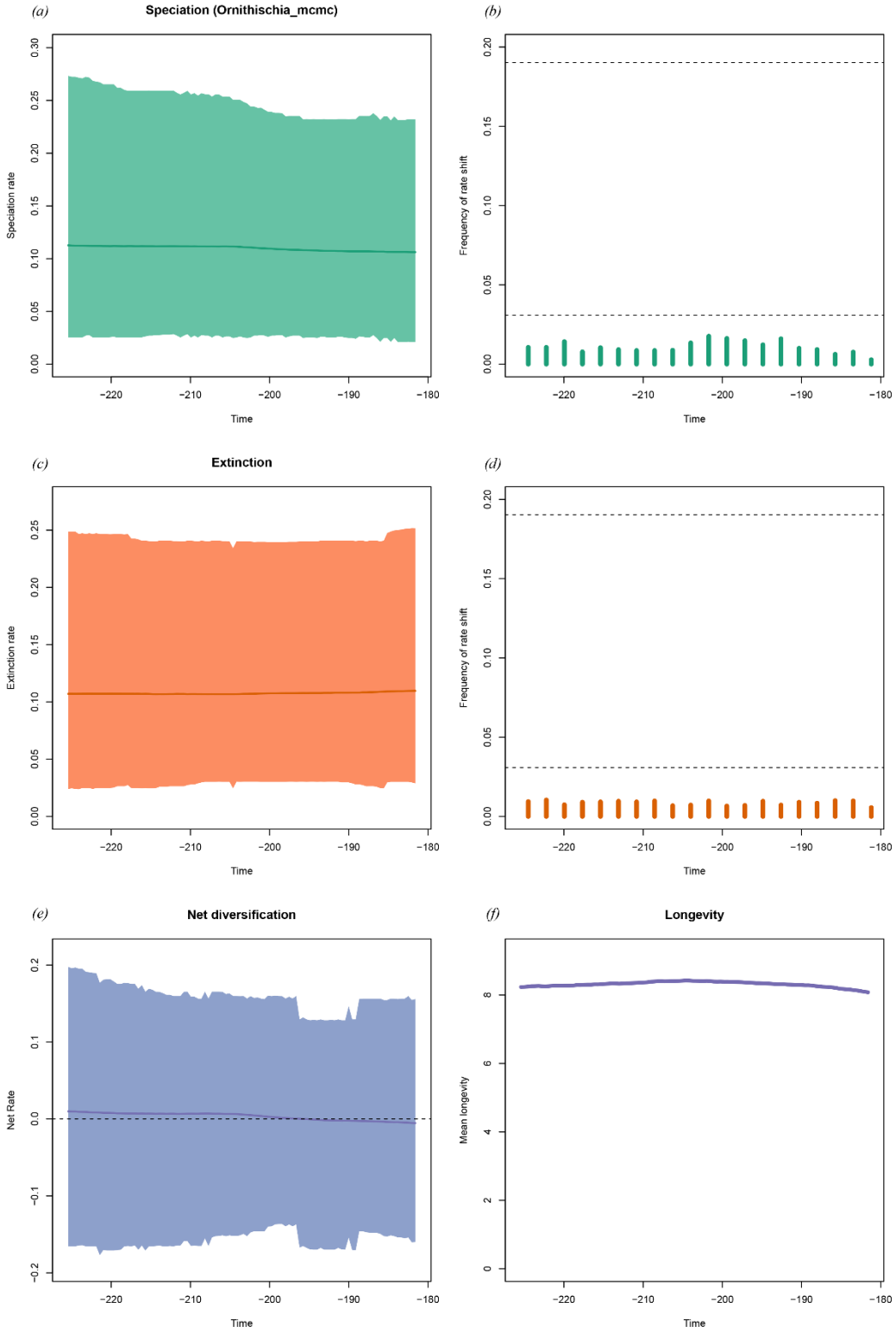


Figure S-11. Full diversification dynamics curves, shifts, and taxon longevities estimated as outputted by PyRate with the HPP model for Ornithischia. Left plots show mean rates and 95% confidence interval, upper two right plots show frequencies of significant rate shifts, and lower right plot shows mean longevity of taxa through time. Speciation (green) and extinction (red) rates are per lineage per Myr, and the diversification rate (purple) is the difference between speciation and extinction rates.

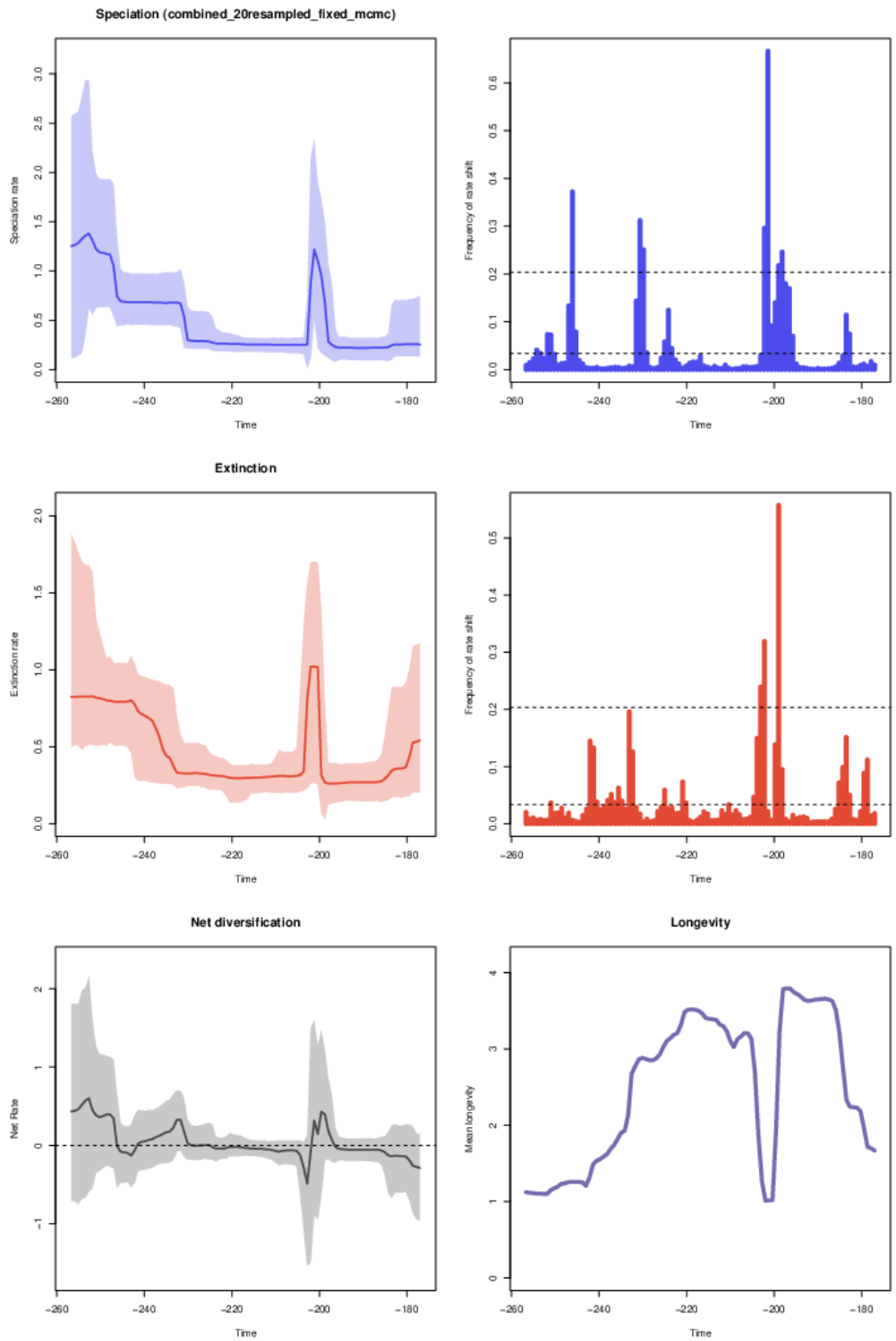


Figure S-12. Full diversification dynamics curves, shifts, and taxon longevities estimated as outputted by PyRate with the HPP model for Archosauromorpha in sensitivity analysis. Left plots show mean rates and 95% confidence interval, upper two right plots show frequencies of significant rate shifts, and lower right plot shows mean longevity of taxa through time. Speciation (green) and extinction (red) rates are per lineage per Myr, and the diversification rate (purple) is the difference between speciation and extinction rates.